Business Experimentation and Causal Methods

Assignment 2: Rocket Fuel

Due Date: 8am, February 6th, 2025

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1. ATE and statistical significance.

1.a What is the $\widehat{ATE}$ of the ads on purchases (conversions)? What does the number mean in words?

In [6]:
```python
ads_data['test'].astype(float)
ads_data['converted'].astype(float)
```

Out[6]:

	converted
0	0.0
1	0.0
2	0.0
3	0.0
4	0.0
...	...
588106	0.0
588107	0.0
588108	0.0
588109	0.0
588110	0.0

588111 rows × 1 columns

dtype: float64

In [7]:
```python
# your code here

a = ads_data[ads_data['test']== 1]['converted'].mean()
b = ads_data[ads_data['test']== 0]['converted'].mean()

ATE = a-b
print(ATE)
```

```
0.007692804899280245
```

The advertising campaign increases the conversion rate by about 0.77 percentage points, which means that compared with those not exposed to the advertisement, users with advertisements are 0.77% more likely to convert.

1.b Did the campaign cause more purchases? Is this difference statistically significant?

Hint: Use the pg.ttest function.

In [9]:
```python
import pingouin as pg
t_test = pg.ttest(ads_data[ads_data['test']== 1]['converted'],ads_data[ads_data['test']== 0]['converted'], pairec
print(t_test)

alpha = 0.05
tstat = t_test.iloc[0]['T']
pvalue = t_test.iloc[0]['p-val']
reject = pvalue < alpha
ci= t_test.iloc[0]['CI95%']
CI_lower = ci[0]
CI_upper = ci[1]

print(f't-statistic(t):{tstat}')
print(f'p-value(p):{pvalue}')
print(f'Do we reject H0? {reject}')
print(f'Confident Interval: {CI_lower}, {CI_upper}')
```

```
              T           dof alternative         p-val                 CI95%  \
T-test  8.65791  26385.467637   two-sided  5.074303e-18  [0.005951, 0.009434]

         cohen-d       BF10  power
T-test  0.049048  1.409e+14    1.0
t-statistic(t):8.657910061879496
p-value(p):5.0743031646645544e-18
Do we reject H0? True
Confident Interval: 0.005951, 0.009434
```

Yes, based on the result of the function, we can see pvalue is very small, which means that if there is no effect, the probability for us to observe the current outcome is extremely small, in other words, the advertisement has effect. What's more, the 95% confidence interval for the difference in conversion rates is [0.005951, 0.009434], which does not include 0. This further supports the conclusion that the advertisement positively impacted purchases.

2. Was the campaign profitable?

2.a How much more profit did TaskaBella make **per person** by showing ads (excluding advertising costs)?

Hint: the profit per conversion is given on page 2 of the case.

[11]
```
noad_conversion = ads_data[ads_data['test']==0]['converted'].mean()
isad_conversion = ads_data[ads_data['test']==1]['converted'].mean()

conversion_diff = isad_conversion - noad_conversion
profit_per_conversion = 40 * conversion_diff
print(profit_per_conversion)
```

```
0.3077121959712098
```

[12]
```
conversion_diff = isad_conversion - noad_conversion
conversion_diff
```

```
0.007692804899280245
```

TaskaBella will make more profits of $0.31 per person by showing ads.

2.b What was the cost of the campaign per person (including the control group)?

Hint: The cost per thousand impressions is $9

In [13]:
```
total_cost = 9 * ads_data['tot_impr'].sum()/1000
cost_per_person = total_cost/ads_data['user_id'].count()
print(cost_per_person)
```

```
0.22340694868825783
```

In [14]:
```
total_cost
```

Out[14]:
```
131388.084
```

The cost of the campaign per person is $0.22.

2.c Calculate the ROI of the campaign (including the control group). Was the campaign profitable?

The ROI is calculated by

$$\mathrm{ROI} = \frac{\text{Effect on Profits per Person in Campaign} - \text{Cost of Ads per Person in Campaign}}{\text{Cost of Ads per Person in Campaign}}$$

In [15]:
```
ROI = (profit_per_conversion - cost_per_person)/cost_per_person
print(ROI)
```

```
0.37736179549451504
```

Yes, the campagin is profitable with a ROI is 0.38

2.d What was the opportunity cost of including a control group --- how much more could TaskaBella have made by not having a control group at all?

Hint: The opportunity cost is the profits that TaskaBella missed out by having a control group.

In [16]:
```
test_conversion_rate = ads_data[ads_data['test']== 1]['converted'].mean()
control_conversion_rate = ads_data[ads_data['test']== 0]['converted'].mean()

conversion_lost_rate = test_conversion_rate - control_conversion_rate
control_number = ads_data[ads_data['test']== 0]['user_id'].count()
opportunity_cost = 40 * conversion_lost_rate * control_number
print(opportunity_cost)
```

```
7238.92941022271
```

The opportunity cost of including a control group is 7238.93, which means that if all member are in the test group, TaskaBella can earn more 7238.93 dollars.

3a. Plot the conversion rate by treatment group and by the number of impressions (groups into ranges, e.g. [0, 100), [100, 500), etc) seen by users.

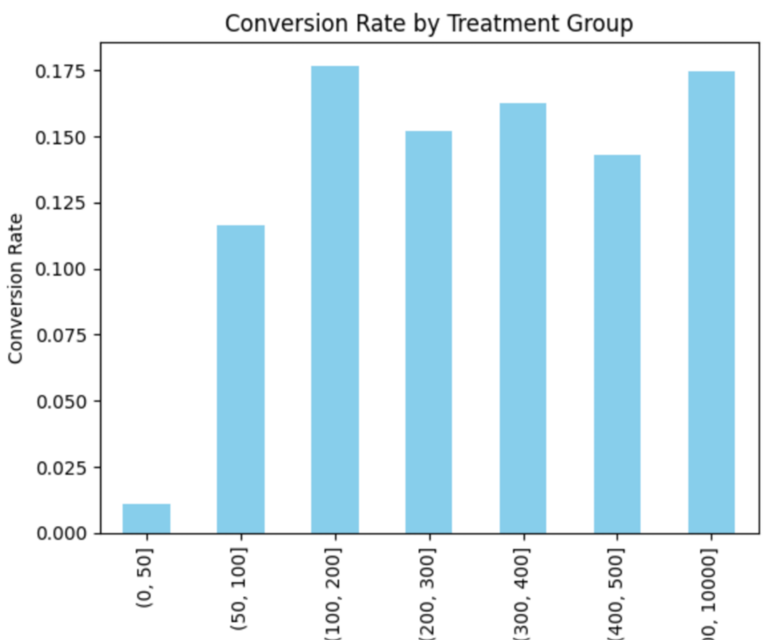
In [17]:
```
bins = pd.IntervalIndex.from_tuples([(0,50), (50,100), (100, 200), (200, 300), (300, 400), (400, 500), (500, 1000
ads_data_1 = ads_data[ads_data['test']==1]
ads_data_1['test_group_impr_count'] = pd.cut(ads_data_1['tot_impr'], bins)
```

```
<ipython-input-17-cc157f0aa831>:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#return
ing-a-view-versus-a-copy
  ads_data_1['test_group_impr_count'] = pd.cut(ads_data_1['tot_impr'], bins)
```

In [18]:
```
import matplotlib.pyplot as plt
conversion_rate = ads_data_1.groupby('test_group_impr_count')['converted'].mean()
conversion_rate.plot(kind='bar', color='skyblue')
plt.title('Conversion Rate by Treatment Group')
plt.xlabel('Impression Bins')
plt.ylabel('Conversion Rate')
plt.show()
```

```
<ipython-input-18-6e50babe1e4d>:2: FutureWarning: The default of observed=False is deprecated and will be changed
to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt th
e future default and silence this warning.
  conversion_rate = ads_data_1.groupby('test_group_impr_count')['converted'].mean()
```



3.b Based on the above figure, can we say that more impressions cause more conversions? (No more than 2 sentences)

No, while the conversion rate shows a strong correlation with impressions (more impressions suggest higher interest, then more likely to order), this does not imply causation, as other factors may influence conversions.

4 Calculate the power of this experiment.

4.a Calculate Cohen's D without using `pg.ttest`. Cohen's D, in this case, is the estimated average treatment effect on conversion divided by the standard deviation of conversion.

In [19]:
```
cd = ATE/ads_data['converted'].std()
print(cd)
```

```
0.04904604110542456
```

The Cohen's D is 0.049

4.b Use the `power_ttest2n` function in `pingouin` to calculate the statistical power of the experiment.

In [20]:
```
n = pg.power_ttest(d=cd, power=.8, alpha=.05,alternative='two-sided')
print(f'Necessary sample size: {np.ceil(n)}')
```

```
Necessary sample size: 6527.0
```

In [21]:
```
power = pg.power_ttest(d=cd, n=6527.0, alpha=0.05, alternative='two-sided')
print(f'Statistic power: {np.round(power, 3)}')
```

```
Statistic power: 0.8
```

4.c What would the power be instead if the true effect had a cohen's D of .01?

In [22]:
```
power_2 = pg.power_ttest(d=.01, n=6527.0, alpha=.05, alternative = 'two-sided')
print(f'Statistic power with cohens d of .01: {np.round(power_2,3)}')
```

```
Statistic power with cohens d of .01: 0.088
```

4.d What would the power be instead if the true effect had a cohen's of .01 and the sample was equally split between treatment and control?

In [23]:
```
power_3 = pg.power_ttest(d=.01, n=ads_data[ads_data['test']==1]['user_id'].count(), alpha=.05, alternative = 'two
print(f'Statistic power with sample splited: {np.round(power_3,3)}')
```

```
Statistic power with sample splited: 1.0
```

5. Case Discussion in Class

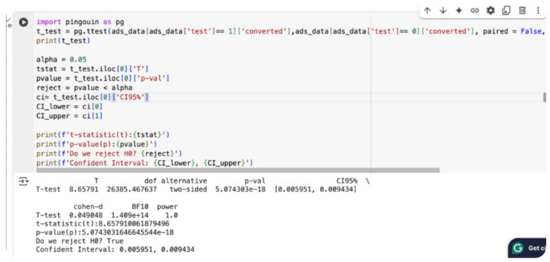
Please write what you would discuss in your presentation to TaskaBella. Your answer should be one paragraph and the paragraph should be five or fewer sentences. Be prepared to discuss in class. Think about what is the most important thing to say to TaskaBella. No additional analysis is needed to answer this question.

Overall, this advertising campaign covered around 590,000 users with 14.5 million impressions in total, which achieved an approximately 2.5% conversion rate with 15,000 users bringing the new handbags. Specifically, the treatment group and control group were exposed to the experiment 24 times per person, but the conversion rate of the treatment group was higher, like 0.77% than the control group. The campaign incurred a total cost of 131,388, averaging 0.22 per person, and generated an ROI of $0.38 per person, reflecting a revenue growth of approximately 172.73% due to the experiment effect. Furthermore, Friday was identified as the peak day of the week, with 11 a.m. to 3 p.m. being the most effective time window of the day.

How long did this problem set take you in hours? How did you find the level of difficulty?

Basic statement: It took me about 4 hours to done, I found it somewhat challenging because it required understanding multiple new concepts and figuring out how they could be calculated, what's more I also spend a lot of time to understand the code that professor had showed on the slides.

GenAI disclosure: I use artificial intelligence to help understand concepts and check code errors. In the part where artificial intelligence is used in the assignment, I pasted the record of my conversation with GPT under the corresponding answer.



it's wired, why is the result of rejection True? but p-value is 5 which is larger than alpha

The result might seem confusing at first glance because of how the `p-value` is displayed. In the result, the `p-value` is **5.074303e-18**, which is in scientific notation and not equal to 5. This actually means:

$$p\text{-value} = 5.074303 \times 10^{-18} = 0.000000000000000005074303$$

how can I use bins to plot a chart, can you give me an example

Certainly! To use the `bins` to plot a chart, you can categorize your data using `pd.cut()` and then create a chart (e.g., a bar chart) to visualize the distribution of data across the bins.

Example: Plotting a Histogram with Bins

Let's say you have some data, and you want to group it into intervals defined by your `bins` and plot the distribution.

python 复制 编辑

```python
import pandas as pd
import matplotlib.pyplot as plt

# Sample data
data = [10, 20, 55, 60, 150, 250, 350, 450, 600, 1000, 2000]

# Define bins
bins = pd.IntervalIndex.from_tuples([(0, 50), (50, 100), (100, 200), (200, 300), (

# Use pd.cut to categorize data into bins
categories = pd.cut(data, bins)
```